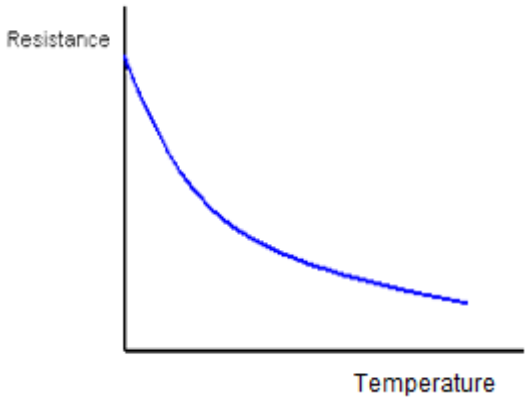


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)			1			1		1
				or alternative graph of resistance increasing with temperature						
		(ii)		thermistor resistance decreases (1) total resistance decreases (1) So ammeter reading increases (1) So voltage across R increases (1) Therefore voltmeter reading decreases (1) Allow converse statements which are linked to candidates graph If alternative graph is used – accept the following thermistor resistance increases (1) total resistance increases(1) So ammeter reading decrease (1) So voltage across R decrease (1) Therefore voltmeter reading increase (1) Answer should link to candidate graph	1	4		5		4

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Light intensity / distance of the torch from the LDR	1			1		1
		(ii)		Resistance of the LDR accept voltmeter reading/ ammeter reading /meter readings	1			1		1
		(iii)		Total resistance = $2.5 + 1 = 3.5 \text{ k}\Omega$ (1) (sub) $I = \frac{9}{3500}$ (1)(manip and conversion) $= 2.57 \times 10^{-3} / 0.00257$ (1) $V = 0.00257 \text{ (ecf)} \times 2500$ (1) (sub) $= 6.42 \text{ V}$ (1) Or Total resistance = $2.5 + 1 = 3.5 \text{ k}\Omega$ (1) (sub) PD across LDR + $1 \text{ kohm} = 9 \text{ V}$ (1) PD across LDR $2.5/3.5 \times 9 = 6.43 \text{ V}$ (1) $I = \frac{6.43 \text{ (ecf)}}{2500}$ (1) (manip and conversion) $= 0.00257$ (1) If conversion not done, expect answers of 2.57 A and $6428/9 \text{ V}$ for (4) Do not penalise incorrect rounding	2	3		5	5	5
		(iv)		Keep light intensity constant (1) Add a variable resistor / use variable power supply (1) Take positive and negative readings of current and voltage (1)			3	3		3
				Question 7 total	6	7	3	16	4	14